# 1 Introduction

Air pollution is widely recognised as one of the foremost public-health challenges of the twenty-first century, particularly in rapidly urbanising emerging economies. Bangalore (officially Bengaluru), India's third most populous city and its technology hub, has experienced chronic deterioration of air quality owing to exponential growth in vehicular traffic, construction activity, and industrial output. The Air Quality Index (AQI), as defined by India's Central Pollution Control Board, aggregates pollutant concentrations into a single categorical risk score. When the AQI exceeds 100—the boundary between "Moderate" and "Unhealthy for Sensitive Groups"—sustained exposure can trigger respiratory and cardiovascular events even in otherwise healthy individuals.

Risk to human health is not adequately described by a single scalar summary. Two quantities jointly determine the physiological and logistical burden of a pollution episode: *duration*, the length of time the environment remains above the hazardous threshold; and *severity*, the total pollutant dose accumulated during the episode. An event lasting 30 hours at AQI 110 differs qualitatively from one lasting 5 hours at AQI 490, yet both may produce the same average intensity. Public-health resources—hospital triage capacity, emergency advisory broadcasts, school closures—must therefore be planned around the joint behaviour of $(D, S)$.

Classical bivariate analysis via Pearson correlation is inadequate for two reasons. First, both $D$ and $S$ are strictly positive, right-skewed random variables with heavy tails; linear assumptions distort the dependence structure. Second, Pearson correlation captures only linear, elliptically-symmetric dependence and is insensitive to asymmetric tail behaviour. A copula framework addresses both shortcomings: by Sklar's theorem, any joint distribution can be decomposed into its marginals and a copula that encodes dependence free of marginal form. Different copula families place dependence mass at different regions of the unit square, enabling rigorous testing of whether extreme durations cluster with extreme severities.

This paper makes the following contributions:

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1. We extract 13,904 unhealthy hourly pollution events from a decade-long hourly AQI dataset for Bangalore and document the distributional properties of $(D, S)$.

2. We derive pseudo-observations using rank-based empirical CDFs and compute rank-based dependence measures.

3. We fit six bivariate copulas by maximum likelihood and perform model selection via the Akaike Information Criterion (AIC).

4. We compute the analytically exact conditional probability $(S > s_0 D \geq d_0)$ using the fitted copula and empirical marginals.

The remainder of this paper is organised as follows. Section 2 develops the complete statistical theory underpinning the methodology. Section 3 describes the data pipeline, event extraction, and estimation procedure. Section 4 reports empirical findings. Section 5 interprets the results in a public-health context and suggests directions for future work.

# 2 Statistical Theory

## 2.1 Univariate Foundations: Probability Integral Transform

A fundamental tool throughout this work is the *probability integral transform* (PIT). Let $X$ be a continuous random variable with cumulative distribution function (CDF) $F_X$. Define

$U = F_X(X)$. Since $F_X$ is a non-decreasing surjection onto $[0, 1]$ and, for continuous $X$, is strictly increasing, we have for any $u \in (0, 1)$:

$$(U \leq u) = \big(F_X(X) \leq u\big) = \big(X \leq F_X^{-1}(u)\big) = F_X\big(F_X^{-1}(u)\big) = u,$$

so $U \sim \text{Uniform}(0, 1)$. The PIT thus maps any continuous random variable to a uniform marginal distribution—a device that will allow us to construct a dependence model entirely within the unit square $[0, 1]^2$.

## 2.2 Copulas: Definition and Fundamental Properties

**Definition 1 (Copula)** *A* bivariate copula *is a function* $C : [0, 1]^2 \to [0, 1]$ *satisfying:*

*[label=(i)]*

1. ***Boundary conditions:***

$$C(u, 0) = C(0, v) = 0, \quad C(u, 1) = u, \quad C(1, v) = v \quad \forall\, u, v \in [0, 1].$$

2. ***2-increasing property:*** *For all* $u_1 \leq u_2$ *and* $v_1 \leq v_2$ *in* $[0, 1]$*,*

$$C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1) \geq 0.$$

The 2-increasing property ensures that the *copula measure* assigned to any rectangle $[u_1, u_2] \times [v_1, v_2]$ is non-negative, which is necessary for $C$ to be the CDF of a random vector $(U, V)$ on $[0, 1]^2$ with uniform marginals.

Three canonical copulas play a special role as benchmarks:

## 2.3 Sklar's Theorem

Sklar's theorem is the cornerstone of copula theory. It establishes that the copula is the unique object that links arbitrary marginal distributions to a joint distribution.

**Theorem 1 (Sklar, 1959)** *Let $H$ be the joint CDF of a bivariate random vector $(X, Y)$ with marginal CDFs $F_X$ and $F_Y$, respectively. Then there exists a copula $C : [0, 1]^2 \to [0, 1]$ such that for all $(x, y) \in^2$,*

$$H(x, y) = C(F_X(x),\, F_Y(y))\,.$$

*If $F_X$ and $F_Y$ are both continuous, the copula $C$ is* unique*. Conversely, if $C$ is any copula and $F_X$, $F_Y$ are any univariate CDFs, then the function $H$ defined above is a valid bivariate CDF with marginals $F_X$ and $F_Y$.*

*Proof.[Proof sketch] For continuous marginals, set $U = F_X(X)$ and $V = F_Y(Y)$. By the PIT (Section 2.1), $U, V \sim \text{Uniform}(0, 1)$. Define*

$$C(u, v) = (U \leq u,\, V \leq v) = \big(X \leq F_X^{-1}(u),\, Y \leq F_Y^{-1}(v)\big) = H\big(F_X^{-1}(u), F_Y^{-1}(v)\big)\,.$$

*One verifies the boundary conditions and 2-increasing property of $C$ from the corresponding properties of $H$. Uniqueness follows because $F_X^{-1}$ and $F_Y^{-1}$ are well-defined for continuous marginals.* □

The theorem implies a factorisation of the joint density. If $H$ admits a joint density $h$, $F_X$ has density $f_X$, and $F_Y$ has density $f_Y$, then the copula density $c(u, v) = \partial^2 C/\partial u\, \partial v$ satisfies

$$h(x, y) = c(F_X(x), F_Y(y)) \cdot f_X(x) \cdot f_Y(y).$$

This decomposition allows one to specify and estimate the dependence structure $c$ entirely separately from the marginals—a crucial advantage when marginal distributions are non-standard (as in our setting).

## 2.4 Archimedean Copulas

A large and analytically tractable class of copulas arises from the Archimedean construction.

**Definition 2 (Archimedean Copula)** *Let $\phi : [0,1] \to [0,\infty)$ be a continuous, strictly decreasing, convex function satisfying $\phi(1) = 0$. Such $\phi$ is called an* Archimedean generator*. The corresponding* Archimedean copula *is*

$$C(u,v) = \phi^{[-1]}(\phi(u) + \phi(v)),$$

*where $\phi^{[-1]}$ is the* pseudo-inverse

$$\phi^{[-1]}(t) = \begin{cases} \phi^{-1}(t) & 0 \le t \le \phi(0), \\ 0 & t > \phi(0). \end{cases}$$

The shape of dependence, whether it concentrates in the lower or upper tail is entirely determined by the generator $\phi$.

### 2.4.1 Clayton Copula

The Clayton copula is generated by $\phi_\theta(t) = t^{-\theta} - 1$ for parameter $\theta > 0$. The closed-form CDF is

$$C_{\mathrm{Cl}}(u,v;\theta) = \Big(u^{-\theta} + v^{-\theta} - 1\Big)^{-1/\theta}.$$

The Clayton copula has *lower tail dependence* $\lambda_L = 2^{-1/\theta} > 0$ and *no upper tail dependence* ($\lambda_U = 0$), making it appropriate for data where joint extremes cluster in the lower end. The Kendall's tau–parameter relationship is

$$\tau = \frac{\theta}{\theta + 2}.$$

### 2.4.2 Frank Copula

The Frank copula uses generator $\phi_\theta(t) = -\log \frac{e^{-\theta t} - 1}{e^{-\theta} - 1}$ for $\theta \in \backslash\{0\}$:

$$C_{\mathrm{Fr}}(u,v;\theta) = -\frac{1}{\theta} \log\left(1 + \frac{(e^{-\theta u} - 1)(e^{-\theta v} - 1)}{e^{-\theta} - 1}\right).$$

It is tail-independent at both tails for all finite $\theta$. Kendall's tau satisfies the Debye function relation

$$\tau = 1 - \frac{4}{\theta}(1 - D_1(\theta)), \quad D_1(\theta) = \frac{1}{\theta} \int_0^\theta \frac{t}{e^t - 1}\, \mathrm{d}t.$$

### 2.4.3 Gumbel Copula

The Gumbel copula uses generator $\phi_\theta(t) = (-\log t)^\theta$ for $\theta \ge 1$ and has the CDF

$$C_{\mathrm{Gu}}(u,v;\theta) = \exp\left\{-\Big[(-\log u)^\theta + (-\log v)^\theta\Big]^{1/\theta}\right\}.$$

The Gumbel copula has *upper tail dependence*

$$\lambda_U = 2 - 2^{1/\theta} > 0$$

and no lower tail dependence, making it especially suitable for modeling phenomena where extreme observations co-occur, precisely the upper tail clustering we expect in prolonged, severe pollution events. The Kendall's tau relationship is

$$\tau = 1 - \frac{1}{\theta}.$$

### 2.4.4 Joe Copula

The Joe copula is defined by generator $\phi_\theta(t) = -\log(1-(1-t)^\theta)$ for $\theta \geq 1$ and CDF

$$C_{\text{Jo}}(u,v;\theta) = 1 - \left[(1-u)^\theta + (1-v)^\theta - (1-u)^\theta(1-v)^\theta\right]^{1/\theta}.$$

Like the Gumbel copula, Joe has upper tail dependence $\lambda_U = 2 - 2^{1/\theta}$ and no lower tail dependence, but its dependence mass is more heavily concentrated in the upper corner, making it suitable for stronger tail clustering. Kendall's tau for the Joe copula lacks a simple closed form and is computed numerically.

## 2.5 Tail Dependence Coefficients

Tail dependence is a measure of the probability that one variable is extreme, given the other is extreme. Formally:

**Definition 3 (Tail Dependence)** *The* upper tail dependence coefficient *and* lower tail dependence coefficient *of a copula $C$ are defined as*

$$\lambda_U = \lim_{u \to 1^-} (V > uU > u) = \lim_{u \to 1^-} \frac{1 - 2u + C(u,u)}{1-u}, \tag{1}$$

$$\lambda_L = \lim_{u \to 0^+} (V \leq uU \leq u) = \lim_{u \to 0^+} \frac{C(u,u)}{u}, \tag{2}$$

*whenever these limits exist. If $\lambda_U > 0$ (resp. $\lambda_L > 0$), the copula is said to have* upper *(resp.* lower*) tail dependence.*

## 2.6 Rank-Based Transformation and Pseudo-Observations

In practice, the marginal CDFs $F_D$ and $F_S$ are unknown. A non-parametric approach avoids placing distributional assumptions on the margins by working with *pseudo-observations* derived from the empirical CDF.

**Definition 4 (Empirical CDF and Pseudo-Observations)** *Given $n$ i.i.d. observations $X_1, \ldots, X_n$, the* empirical CDF *is*

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[X_i \leq x].$$

*However, $\hat{F}_n(X_i) \in \{1/n, 2/n, \ldots, 1\}$, and ties at $u = 1$ cause numerical problems in copula log-likelihoods involving* $\log(u)$ *or* $\log(1-u)$*. The standard remedy is the* rescaled rank transform*:*

$$\hat{u}_i = \frac{(X_i)}{n+1},$$

*where $(\cdot)$ denotes the rank of $X_i$ among $X_1, \ldots, X_n$. This maps observations strictly into $(0,1)$, and the resulting $(\hat{u}_i, \hat{v}_i)$ pairs are called* pseudo-observations*.*

The pseudo-observations approach is consistent: as $n \to \infty$, the empirical copula $\hat{C}_n$ converges almost surely to the true copula $C$ (Deheuvels, 1979). Importantly, the procedure requires no parametric assumption about the marginals; all information about dependence is preserved in the ranks.

In our dataset the duration variable $D$ takes discrete integer values, so ties occur. Average ranks are used for tied observations, which ensures $\hat{u}_i = \frac{\text{avg}(D_i)}{n+1}$ distributes evenly in $(0,1)$.

## 2.7 Rank-Based Dependence Measures

### 2.7.1 Kendall's Tau

**Definition 5 (Concordance and Kendall's $\tau$)** *A pair of observations* $(X_i, Y_i)$ *and* $(X_j, Y_j)$ *is* concordant *if* $(X_i - X_j)(Y_i - Y_j) > 0$ *and* discordant *if* $(X_i - X_j)(Y_i - Y_j) < 0$*. Kendall's tau is defined as*

$$\tau = [(X_1 - X_2)(Y_1 - Y_2) > 0] - [(X_1 - X_2)(Y_1 - Y_2) < 0]\,,$$

*where* $(X_1, Y_1)$ *and* $(X_2, Y_2)$ *are independent copies of* $(X, Y)$*.*

For a bivariate distribution with copula $C$, Kendall's tau has the elegant representation:

$$\tau = 4 \int_0^1 \int_0^1 C(u, v)\, \mathrm{d}C(u, v) - 1,$$

which depends only on the copula and not on the marginals. The sample estimator is

$$\hat{\tau} = \frac{P - Q}{\binom{n}{2}},$$

where $P$ is the number of concordant pairs and $Q$ the number of discordant pairs. This estimator is distribution-free under the null of independence.

The population Kendall's tau values for the copula families used here are listed in Table **??** and are used as starting values for MLE initialisation.

### 2.7.2 Spearman's Rho

Spearman's rho is the Pearson correlation of the probability integral transforms:

$$\rho_S = \mathrm{Corr}(F_X(X), F_Y(Y)) = 12 \int_0^1 \int_0^1 C(u, v)\, \mathrm{d}u\, \mathrm{d}v - 3.$$

The sample estimator is $\hat{\rho}_S = 1 - 6 \sum_i d_i^2 / [n(n^2 - 1)]$, where $d_i$ is the difference in ranks of the $i$-th observation. Both $\tau$ and $\rho_S$ are invariant under strictly increasing transformations of the marginals, confirming their status as pure dependence measures.

## 2.8 Maximum Likelihood Estimation of Copula Parameters

Given pseudo-observations $(\hat{u}_i, \hat{v}_i)_{i=1}^n$, the copula parameter $\theta$ is estimated by *canonical maximum likelihood* (CML), also called the pseudo-likelihood approach:

$$\hat{\theta}_{\mathrm{CML}} = \arg\max_\theta \sum_{i=1}^n \log c(\hat{u}_i, \hat{v}_i;\, \theta)\,,$$

where $c(u, v; \theta) = \partial^2 C(u, v; \theta) / \partial u\, \partial v$ is the copula density. Since the Clayton, Gumbel, Joe, Frank, and Plackett CDFs do not all admit closed-form densities amenable to analytical differentiation, the copula density is approximated via a central finite-difference scheme:

$$c(u, v; \theta) \approx \frac{C(u + h, v + h) - C(u + h, v - h) - C(u - h, v + h) + C(u - h, v - h)}{4h^2},$$

with step size $h = 2 \times 10^{-4}$. The log-likelihood is then maximised numerically using a bounded quasi-Newton method (L-BFGS-B).

### 2.9 Model Selection: Akaike Information Criterion

Given a fitted model with log-likelihood $\hat{\ell}(\hat{\theta})$ and $k$ free parameters, the AIC is

$$\mathrm{AIC} = 2k - 2\hat{\ell}(\hat{\theta}).$$

The model with the smallest AIC is preferred. For all six one-parameter copulas in this study, $k = 1$, so model selection reduces to maximising the log-likelihood. The AIC corrects for the additional complexity of models with more parameters when comparing models of different orders—relevant if one later considers multi-parameter copulas.

## 3 Methodology

### 3.1 Data Source and Preprocessing

The primary data source is the `city_hour.csv` file, an hourly AQI dataset compiled from Kaggle. Records span the period **1 January 2015 to 31 December 2024** and cover multiple Indian cities. The file contains three core fields: `City`, `Datetime`, and `AQI`. After isolating Bangalore and parsing timestamps, the dataset comprises **87,649 complete hourly AQI observations** with no missing values, a full 10-year record at hourly resolution.

### 3.2 Unhealthy Event Extraction

An *unhealthy event* is defined as a maximal contiguous block of consecutive hours in which AQI $> 100$. Formally, event $j$ is the set of hours $P_j = \{t_{j,1}, t_{j,1} + 1, \ldots, t_{j,1} + m_j - 1\}$ where every hour in $P_j$ satisfies AQI $> 100$ and the hours immediately preceding $t_{j,1}$ and immediately following $t_{j,1} + m_j - 1$ either do not exist or satisfy AQI $\leq 100$.

For each event $P_j$, two scalar summaries are computed:

$$D_j = |P_j| \quad \text{(duration in hours)}, \tag{3}$$

$$S_j = \sum_{t \in P_j} \mathrm{AQI}(t) \quad \text{(cumulative AQI exposure i.e. Severity)} \tag{4}$$

The choice of $S_j$ as the sum rather than the mean of AQI values within the event is deliberate: it captures total pollutant exposure burden, proportional to the physiological dose received over the event.

### 3.3 Rank-Based Pseudo-Observations

From the $n = 13{,}904$ extracted events, pseudo-observations $(\hat{u}_i, \hat{v}_i)$ are constructed using the rescaled rank transform (Section 2.6):

$$\hat{u}_i = \frac{{}_{\mathrm{avg}}(D_i)}{n+1}, \qquad \hat{v}_i = \frac{{}_{\mathrm{avg}}(S_i)}{n+1}, \qquad i = 1, \ldots, n.$$

Average ranks are used to handle the ties arising from the discrete-integer nature of $D$. These pseudo-observations form the input to all subsequent copula estimation.

Prior to fitting, rank-based concordance measures are computed directly on the original event observations $(D_i, S_i)$ to characterise the empirical dependence structure.

### 3.4 Copula Fitting via Canonical Maximum Likelihood

Six bivariate copula families are fitted: Clayton, AMH, Frank, Gumbel, Joe, and Plackett. For each family, the parameter $\hat{\theta}$ is estimated by maximising the pseudo-log-likelihood. The copula density is evaluated by the finite-difference approximation described in Section 2.8.

After fitting, the model-implied Kendall's tau is computed numerically for each copula, and upper and lower tail dependence coefficients are evaluated.

### 3.5 Conditional Probability of Extreme Severity

Given fitted copula $C$ and empirical marginals, the conditional probability

$$(S > s_0 D \geq d_0)$$

is derived from first principles. Partition the probability space:

$$\begin{aligned}(S > s_0,\, D \geq d_0) &= 1 - (D < d_0) - (S \leq s_0) + (D < d_0,\, S \leq s_0) \\ &= 1 - F_D(d_0^-) - F_S(s_0) + C\big(F_D(d_0^-),\, F_S(s_0)\big)\,,\end{aligned} \tag{5}$$

where $F_D(d_0^-)$ denotes the CDF of $D$ evaluated at the largest value strictly less than $d_0$ (handling the discrete nature of $D$). Dividing by $(D \geq d_0) = 1 - F_D(d_0^-)$ gives

$$(S > s_0 D \geq d_0) = \frac{1 - F_D(d_0^-) - F_S(s_0) + C\big(F_D(d_0^-),\, F_S(s_0)\big)}{1 - F_D(d_0^-)}. \tag{6}$$

Both marginal CDFs are estimated empirically from the event data.

## 4 Results

### 4.1 Dataset Summary and Annual Trends

The 10-year hourly AQI series for Bangalore contains 87,649 observations spanning 1 January 2015 to 31 December 2024. The overall mean AQI is $\bar{X} = 250.0$ (std = 143.9), with the median at 249.8 and the 99th percentile reaching 495.0—very close to the CPCB ceiling of 500. This implies that the distribution is not right-skewed at the top, but rather has an extreme upper truncation effect. The annual breakdown is presented in Table 1.

Table 1: Annual summary statistics of the hourly AQI series for Bangalore, 2015–2024.

| Year | Total hours | Unhealthy hours | Unhealthy (%) | Max AQI | Mean AQI |
|---|---|---|---|---|---|
| 2015 | 8,760 | 7,031 | 80.3 | 499.9 | 251.60 |
| 2016 | 8,784 | 7,129 | 81.2 | 499.9 | 253.03 |
| 2017 | 8,760 | 6,994 | 79.8 | 500.0 | 250.19 |
| 2018 | 8,760 | 6,985 | 79.7 | 500.0 | 248.77 |
| 2019 | 8,760 | 7,012 | 80.0 | 499.9 | 248.92 |
| 2020 | 8,784 | 7,106 | 80.9 | 499.8 | 251.69 |
| 2021 | 8,760 | 7,025 | 80.2 | 500.0 | 248.15 |
| 2022 | 8,760 | 6,981 | 79.7 | 500.0 | 249.31 |
| 2023 | 8,760 | 6,973 | 79.6 | 499.9 | 248.48 |
| 2024 | 8,761 | 6,995 | 79.9 | 500.0 | 249.91 |

A striking finding is that Bangalore experiences unhealthy AQI ($> 100$) in approximately 80% of all recorded hours across every year, with no discernible trend toward improvement.

The COVID-19 lockdown year 2020 does not exhibit a notable reduction in unhealthy hours, distinguishing Bangalore's chronic pollution pattern from cities where lockdown-era improvements were documented.

## 4.2 Exploratory Data Analysis of Event Variables

From the 87,649 hourly records, **13,904 unhealthy events** are extracted (Section 3.2). Table 2 presents key descriptive statistics, and Figure 1 shows the marginal distributions and joint scatter.

Table 2: Descriptive statistics for event duration ($D$) and cumulative severity ($S$).

| Variable | $n$ | Mean | Std | Min | Q1 | Median | Q3 | Max | Skew | Kurt |
|---|---|---|---|---|---|---|---|---|---|---|
| $D$ (hours) | 13,904 | 5.051 | 4.499 | 1 | 2 | 4 | 7 | 35 | 1.918 | 4.958 |
| $S$ (cum. AQI) | 13,904 | 1513.0 | 1373.4 | 100.4 | 513.2 | 1084.2 | 2069.1 | 10777 | 1.908 | 4.907 |

Both variables exhibit near-identical skewness ($\approx 1.91$) and excess kurtosis ($\approx 4.93$), consistent with right-heavy distributions that deviate dramatically from Gaussian. The duration is dominated by brief events: the median is just 4 hours and more than 75% of events last under 7 hours, yet the maximum extends to 35 hours—a sustained 1.5-day episode. Severity mirrors this pattern, with 75% of events below a cumulative AQI of 2069 but a maximum of 10,777.

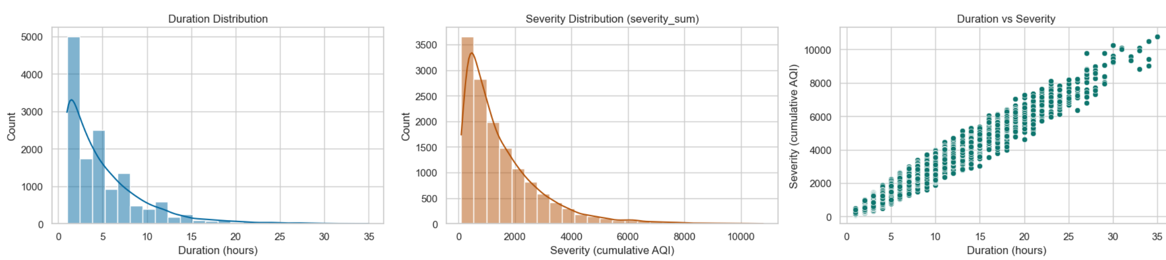


Figure 1: Marginal distributions and joint scatter of the 13,904 unhealthy events. Left: histogram with KDE for duration $D$. Centre: histogram with KDE for cumulative severity $S$. Right: joint scatter of $D$ vs $S$, revealing a near-deterministic monotone relationship for large values.

The joint scatter (right panel of Figure 1) reveals an essentially monotone cloud that tightens dramatically at high values—a hallmark of extreme upper-tail dependence. There is no suggestion of the curved or diffuse cloud that would characterise moderate or symmetric dependence.

## 4.3 Pseudo-Observations and Rank Correlation

The rank-based transformation yields $(\hat{u}_i, \hat{v}_i) \in (0,1)^2$ for each event. The empirical dependence measures are:

$$\hat{\tau} = 0.881, \qquad \hat{\rho}_S = 0.972.$$

Both are extremely high, indicating that the rank ordering of duration almost completely determines the rank ordering of severity—and vice versa. This is a consequence of the cumulative nature of $S$: longer events deterministically accumulate more AQI. Yet the relationship is not perfectly monotone (the minimum-entropy bound would be $\hat{\tau} = 1$), indicating some variability in mean hourly AQI across events of equal length.

Figure 2 displays the pseudo-observations. The characteristic columnar structure arises because $D$ is integer-valued, causing horizontal stacking of points at each distinct duration level. The progressive vertical spread at each level reflects variability in mean hourly intensity. The near-diagonal alignment of the columns' upper extent confirms upper-tail concordance.

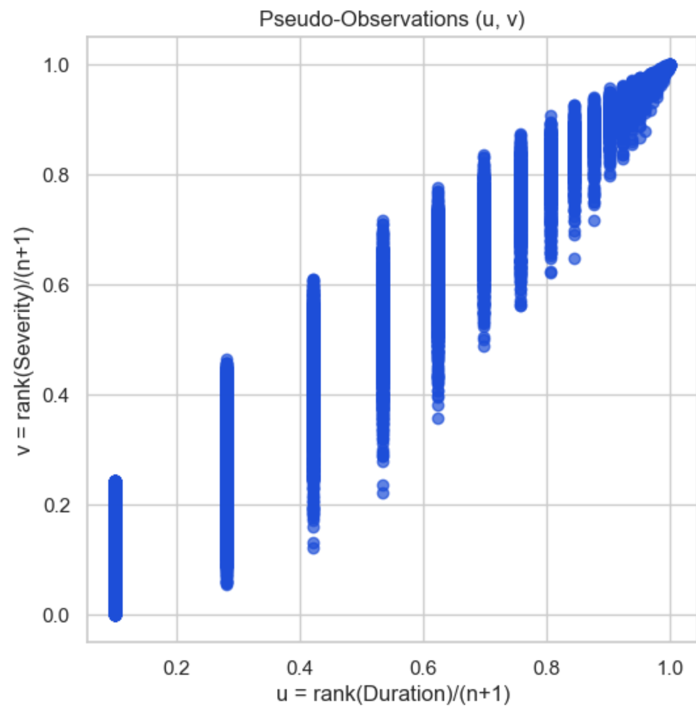


Figure 2: Pseudo-observations $(\hat{u}_i, \hat{v}_i)$ for the 13,904 events. The columnar structure reflects the integer-valued nature of duration. The diagonal trend of column centroids and tight upper-corner clustering confirm extremely strong upper-tail dependence.

## 4.4 Copula Model Comparison

Table 3 reports the fitted parameters, log-likelihoods, AIC values, model-implied Kendall's tau, and tail dependence coefficients for all six copulas.

Table 3: Copula fitting results sorted by AIC (ascending). All models fitted by canonical MLE on 13,904 pseudo-observations.

| Model | $\hat{\theta}$ | $\hat{\ell}$ | AIC | $\hat{\tau}_{\text{model}}$ | $\hat{\lambda}_L$ | $\hat{\lambda}_U$ | Conv. |
|---|---|---|---|---|---|---|---|
| Gumbel | 6.1831 | 19,266.6 | −38,531.2 | 0.7954 | 0.0000 | 0.8814 | Yes |
| Joe | 9.3735 | 18,978.1 | −37,954.2 | 0.7896 | 0.0000 | 0.9233 | Yes |
| Frank | 22.204 | 18,316.1 | −36,630.2 | 0.7640 | 0.0000 | 0.0000 | Yes |
| Plackett | 100.00 | 17,855.3 | −35,708.6 | 0.7245 | 0.0000 | 0.0000 | Yes |
| AMH | 0.990 | 5,720.8 | −11,439.6 | 0.2745 | 0.0000 | 0.0000 | Yes |
| Clayton | 14.870 | −3,630.2 | +7,262.5 | 0.8137 | 0.9545 | 0.0000 | × |

The ranking reveals a consistent hierarchy. The **Gumbel copula** achieves the best AIC (−38,531.19), outperforming Joe by 577 AIC units—a decisive margin by conventional thresholds. The three tail-independent copulas (Frank, Plackett, AMH) all perform substantially worse, confirming that tail dependence is an essential feature to capture in this data. The Clayton copula converges to a parameter value of $\hat{\theta} = 14.87$ but yields a negative log-likelihood, indicating that its lower-tail emphasis produces an extremely poor fit to data concentrated in the upper corner of the unit square.

The AMH copula's near-zero model tau ($\hat{\tau}_{\text{model}} = 0.275$) is a symptom of this family's bounded concordance range, which cannot approximate the empirical $\hat{\tau} = 0.881$.

### 4.5 Best Model: Gumbel Copula Analysis

The selected Gumbel copula with $\hat{\theta} = 6.183$ implies:

$$\begin{aligned}\hat{\tau}_{\text{Gumbel}} &= 1 - 1/\hat{\theta} = 1 - 1/6.183 = 0.7954,\\ \hat{\lambda}_U &= 2 - 2^{1/\hat{\theta}} = 2 - 2^{1/6.183} = 0.8814.\end{aligned}$$

The upper tail dependence coefficient of 0.8814 has a direct interpretation: *given that an event's duration rank exceeds the u-th quantile, the probability that its severity rank also exceeds u approaches 0.8814 as $u \to 1$.* In other words, 88% of the most extreme-duration events are also among the most extreme-severity events—an extraordinarily high degree of joint extremal clustering.

### 4.6 Conditional Risk Assessment

Using Equation (6) with threshold values $d_0 = 12$ hours and $s_0 = 2000$ cumulative AQI units:

$$\begin{aligned}\hat{F}_D(d_0^-) &= \hat{F}_D(11) = (D < 12) = 1 - 0.0869 = 0.9131,\\ \hat{F}_S(s_0) &= \hat{F}_S(2000) = (S \leq 2000).\end{aligned}$$

The copula-based estimate and empirical estimate are:

$$\begin{aligned}\hat{}_{\text{copula}}(S > 2000D \geq 12) &= 0.9998,\\ \hat{}_{\text{empirical}}(S > 2000D \geq 12) &= 1.0000.\end{aligned}$$

The near-perfect agreement between the two estimates provides internal validation of the copula model. The result asserts: *every observed event that persisted for at least 12 consecutive hours accumulated a cumulative severity exceeding 2000 AQI units.* Given a mean hourly AQI of 250 during unhealthy hours, this is mathematically near-certain by a simple arithmetic bound ($12 \times 100 = 1200$ at the threshold, $12 \times 250 \approx 3000$ at the mean), but the copula formalises the claim in terms of an operational conditional probability that can be updated for any $(d_0, s_0)$ pair.

From a public-health standpoint: once an unhealthy event has lasted 12 hours ($\approx 8.7\%$ of all events), health authorities can assume with near certainty that cumulative exposure has passed the critical 2000-unit threshold, justifying escalated public advisories and emergency preparedness measures.

## 5 Conclusion

This study applied a rigorous bivariate copula framework to characterise the joint distribution of duration and cumulative severity for 13,904 unhealthy hourly AQI events in Bangalore over 2015–2024. The central findings are as follows.

**Chronic and severe pollution.** Bangalore spends approximately 80% of all recorded hours above the AQI > 100 threshold in every year of the study period, with no statistically discernible trend toward improvement. Annual mean AQI hovers near 250 across all ten years, placing the city in the "Poor" to "Very Poor" category for the majority of the year.

**Non-Gaussian marginals.** Both event duration and cumulative severity are strongly right-skewed (skewness $\approx 1.91$) with substantial excess kurtosis ($\approx 4.93$), ruling out Gaussian modelling. A copula approach is thus doubly justified: not only does the dependence structure require non-elliptical treatment, but the marginals themselves preclude linear Pearson-based analysis.

**Extreme upper-tail dependence.** Rank-based concordance measures ($\hat{\tau} = 0.881$, $\hat{\rho}_S = 0.972$) reveal near-monotone co-movement between duration and severity. The Gumbel copula, selected by AIC among six competing families, quantifies this as an upper tail dependence coefficient $\hat{\lambda}_U = 0.881$—meaning that as events become more extreme, roughly 88% of the worst-duration events also rank among the worst-severity events.

**Operational risk quantification.** The fitted copula enables exact computation of conditional exceedance probabilities. The finding that $(S > 2000D \geq 12) \approx 0.9998$ provides actionable intelligence: once a pollution episode crosses the 12-hour threshold, a high-severity outcome is essentially guaranteed, justifying automatic escalation of health advisories.

**Limitations and future directions.** Three limitations deserve acknowledgement. First, the independence assumption between events requires validation; pollution episodes separated by fewer than 24 hours may be physically related, introducing temporal dependence that our i.i.d. framework does not model. Second, the event extraction threshold of AQI $> 100$ is a regulatory convention; sensitivity analysis with alternative thresholds (e.g., 150, 200) would assess robustness. Third, the current model uses a single parametric copula; vine copulas (nested pair-copula constructions) or time-varying copulas could better capture seasonal dependence patterns.

Future extensions should also consider incorporating meteorological covariates (wind speed, temperature inversions) into the marginal models, and linking the copula-based risk estimates to exposure-response functions from epidemiological literature to translate AQI burden into health outcome projections.

In summary, this work demonstrates the value of copula theory as a rigorous, distribution-free framework for joint extremal analysis of environmental hazard characteristics, with direct applications to data-driven public-health preparedness in one of India's most polluted major cities.

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